KLE Technological University
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Research Experience for Undergraduates

A Research Project on

Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

Research Project Report submitted in partial fulfilment of 7th semester requirement for

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in
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By

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CERTIFICATE

This is to certify that research on "*Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring"* is a Bonafide work carried out by the student *Rohan Kulkarni*, SRN: *02FE22BCS083* in partial fulfilment of the award of Degree of Bachelor of Engineering in Computer Science and Engineering during the year 2025 - 26. The research thesis has been approved as it satisfies the academic requirement with respect to the research work prescribed for the above said program.

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Abstract

Intrusion Detection Systems (IDS) are essential for protecting IoT networks; however, many deep learning–based IDS solutions are too computationally intensive to be deployed on resource-constrained edge devices. This paper proposes TSANet, a lightweight Temporal–Spatial Attention Network for real-time intrusion detection on low-power hardware such as the Raspberry Pi 5. The proposed model integrates a 1D Convolutional Neural Network for spatial feature learning, an LSTM network for tempo ral sequence modeling, and attention mechanisms to highlight important traffic features and time steps. TSANet is trained and evaluated on the CICIDS2017 dataset and optimized using pruning and quantization-aware training to reduce model size and inference latency. Experimental results show that the optimized model achieves 98.3% offline accuracy and 96.8% real-time detection accuracy on Raspberry Pi, with an inference latency of 8–12 ms per flow, demonstrating its effectiveness for practical IoT edge based intrusion detection.

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Chapter 1

Introduction

This work aligns with the United Nations Sustainable Development Goal 9 (SDG 9), which emphasizes building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation through secure and reliable digital systems. The rapid growth of Internet of Things (IoT) technologies has led to the large-scale deployment of connected devices in smart homes, healthcare systems, industrial automation, and smart cities. These devices continuously exchange data over networks and often operate in open or untrusted environments, making them attractive targets for cyberattacks such as denial of-service (DoS), distributed denial-of-service (DDoS), malware injection, and data breaches. Ensuring the security of IoT networks has therefore become a critical challenge

Intrusion Detection Systems (IDS) are widely used to monitor network traffic and identify malicious activities. Traditional IDS approaches rely on signature based or rule-based techniques, which are effective for known attacks but fail to detect novel or evolving threats. In recent years, deep learning–based IDS solutions have demonstrated strong detection performance by learning complex patterns in network traffic.

However, most existing deep learning models are computationally heavy and designed to run on powerful servers or cloud infrastructure. Deploying such models directly on IoT edge devices is difficult due to limited processing power, memory, and energy availability. Transmitting raw network data to centralized servers also introduces latency and raises privacy concerns. These limitations highlight the need for lightweight, real-time intrusion detection systems that can operate directly on edge devices without relying on ex ternal computation.

To address these challenges, this work proposes TSANet, a lightweight Tem poral–Spatial Attention Network designed for efficient deployment on resource constrained hardware. The proposed system performs complete intrusion detection at the edge, including traffic analysis and real-time inference, and is vali dated through deployment on a Raspberry Pi 5 using both benchmark datasets and live network traffic. The results demonstrate that effective intrusion detection can be achieved at the IoT edge with low latency and high accuracy.

1.1 Background

The rapid expansion of Internet of Things (IoT) ecosystems across smart homes, industrial automation, healthcare, and critical infrastructure has significantly increased the attack surface of modern networks. IoT devices often operate with limited computational power, memory, and energy resources, making them particularly vulnerable to cyber-attacks such as Distributed Denial of Service (DDoS), botnet infections, scanning, and protocol exploitation. Traditional network security mechanisms, which rely on centralized monitoring and heavyweight intrusion detection systems (IDS), are often unsuitable for such resource-constrained environments.

Conventional intrusion detection systems primarily depend on signature-based or rule-based techniques, which require prior knowledge of attack patterns and frequent manual updates. While effective against known threats, these approaches struggle to detect zero-day attacks and rapidly evolving adversarial behaviors commonly observed in IoT networks. Machine learning–based IDS solutions improved adaptability by learning statistical patterns from network traffic; however, their performance is heavily dependent on handcrafted features and labeled datasets, limiting robustness in dynamic real-world scenarios.

Recent advances in deep learning have enabled intrusion detection models to automatically learn complex spatial and temporal patterns from network traffic. Architectures such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks have demonstrated superior detection accuracy compared to classical methods. Nevertheless, most deep learning–based IDS models are designed for cloud or server-grade infrastructure, requiring high computational resources and incurring latency due to centralized processing. This makes them impractical for real-time deployment at the IoT edge, where low latency, data privacy, and autonomous operation are critical.

1.2 Problem Statement

IoT networks generate continuous, diverse, and often unpredictable traffic, yet most existing Intrusion Detection Systems (IDS) rely on heavy deep-learning models that cannot run on resource-constrained edge devices like the Raspberry Pi. Centralized cloud-based IDS solutions introduce latency, privacy risks, and bandwidth overhead, while many lightweight IDS approaches fail to capture the temporal and spatial behaviour of real attacks. Moreover, most research is evaluated only on offline datasets and not on live Wi-Fi traffic, making them impractical for real deployments

1.2.1 Objectives

The objectives of this project are as follows:

- **To design a lightweight deep learning–based intrusion detection system for IoT networks:**
  - Develop an efficient intrusion detection model capable of identifying malicious network activities while operating under the computational constraints of IoT edge devices.
- **To implement a Temporal–Spatial Attention Network (TSANet) for real-time anomaly detection:**
  - Integrate convolutional, recurrent, and attention mechanisms to effectively capture both spatial feature dependencies and temporal traffic patterns in network flows.
- **To optimize the proposed model for edge deployment:**
  - Apply model optimization techniques such as pruning and quantization-aware training to reduce inference latency and memory usage without compromising detection accuracy.
- **To deploy and validate the intrusion detection system on Raspberry Pi hardware:**
  - Configure a Raspberry Pi as a live network gateway and deploy the trained model to perform real-time traffic capture, feature extraction, and on-device inference.

Chapter 2

Literature Survey

Intrusion detection in IoT networks has been widely studied due to the rapid growth of connected devices and the increasing frequency of cyberattacks. This section reviews key research works relevant to deep learning–based and edge-deployable intrusion detection systems for IoT environments.

2.1 Early and Foundational Works

Shone et al. proposed a deep autoencoder-based intrusion detection system capable of learning latent representations of network traffic, improving the detection of previously unseen attacks **[14]**. While effective, the model was evaluated primarily on benchmark datasets and not designed for edge deployment.

Tsimenidis et al. presented a comprehensive survey on deep learning techniques for IoT intrusion detection, highlighting challenges such as model complexity, data imbalance, and real-time deployment constraints **[11]**. The study emphasized the need for lightweight and deployable IDS solutions.

Morshedi et al. investigated deep learning–based intrusion detection for IoT security, achieving high detection accuracy across multiple datasets **[12].** However, the proposed models lacked evaluation on real-world edge hardware.

2.2 Deep Learning-Based IDS for IoT

Idrissi et al. developed a lightweight optimized deep learning–based host intrusion detection system deployed at the IoT edge **[2]**. Their approach reduced computational overhead while maintaining high detection accuracy, demonstrating the feasibility of deep learning on constrained devices.

Saba et al. proposed an anomaly-based intrusion detection system using deep learning models to detect malicious activities in IoT networks **[3]**. The system showed improved performance compared to traditional methods but relied on offline datasets.

Otoum and Nayak introduced DL-IDS, a deep learning–based intrusion detection framework designed for securing IoT systems **[5]**. Their work highlighted the importance of sequential learning for detecting complex and evolving attacks.

Gupta et al. focused on intrusion detection in 5G-enabled IoT networks using deep learning techniques **[13]**. The study emphasized low-latency detection requirements but did not explore full edge deployment.

2.3 Hybrid and Attention-Based IDS Approaches

Gueriani et al. proposed a hybrid CNN–LSTM intrusion detection system to capture both spatial and temporal characteristics of network traffic **[4]**. The hybrid model improved detection accuracy compared to single-model approaches.

Kayode et al. introduced a Temporal–Spatial Attention Network (TSANet) for detecting DoS attacks in network traffic **[6]**. Their work demonstrated that attention mechanisms enhance feature selection, though deployment was not fully evaluated on IoT edge hardware.

Racherla et al. developed a real-time deep intrusion detection system for IoT nodes, achieving low-latency detection suitable for constrained environments **[7]**. However, the study did not include live traffic experimentation.

2.4 Edge and Raspberry Pi–Based IDS

Dahiya et al. presented a Raspberry Pi–based intrusion detection system for IoT networks **[8]**. The system demonstrated feasibility on low-power hardware but lacked extensive real-time performance evaluation.

Olakanmi and Olatunji proposed an optimized ensemble deep learning IDS deployed on Raspberry Pi devices **[9]**. Their approach reduced resource usage while maintaining accuracy, showing promise for edge-based IDS.

Adrian et al. designed and implemented an adaptive machine learning–based intrusion detection system for smart home networks using Raspberry Pi **[10].**The system adapted to changing traffic patterns but relied on traditional ML techniques

Chapter 3

System Flow

3.1 System Flow

This chapter describes the overall system flow of the proposed lightweight intrusion detection framework designed for real-time IoT network security. The system integrates traffic capture, feature extraction, deep learning–based classification, and alert generation, all executed directly on a resource-constrained edge device. The objective is to detect malicious activities in real time without relying on centralized cloud infrastructure.

3.1.1 Overview of the System Workflow

The proposed system follows an end-to-end edge-based intrusion detection pipeline. First, network traffic is captured directly from live communication when the Raspberry Pi operates as a Wi-Fi access point. The captured packets are then grouped into bidirectional flows, from which relevant numerical features are extracted. These features are preprocessed and passed to the trained TSANet model for classification. Based on the prediction outcome, the system identifies the traffic as benign or malicious and generates alerts locally.

3.1.2 Feature Extraction and Preprocessing

From each generated flow, a set of 32 engineered bidirectional features is extracted. These include packet count statistics, byte-level information, flow duration, and inter-arrival time metrics. The selected features provide sufficient discriminative power while keeping computational overhead low.

The extracted features are then normalized using Min–Max scaling to ensure consistent numerical ranges during inference. Since feature extraction and preprocessing are standard stages in IDS pipelines, they are optimized for fast execution to support real-time operation.

3.1.3 Traffic Capture and Flow Generation

The Raspberry Pi 5 is configured as a wireless access point, allowing multiple client devices such as laptops and mobile phones to connect through it. All incoming and outgoing packets are captured in real time using packet sniffing tools. Instead of analysing individual packets, the system aggregates packets into bidirectional flows defined by source IP, destination IP, source port, destination port, and protocol as we can see from figure 1.

This flow-based approach reduces noise and provides a structured representation of network behavior, which is more effective for intrusion detection than packet-level analysis.

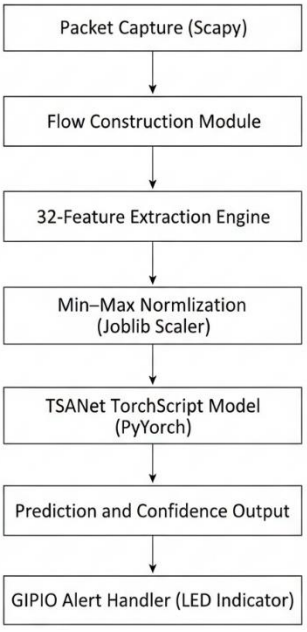

Fig 1: End-to-end system workflow

3.1.4 Intrusion Detection Using TSANet

The preprocessed feature vectors are passed to the Temporal–Spatial Attention Network (TSANet). The model first applies convolutional layers to learn spatial correlations among features, followed by LSTM layers to capture temporal dependencies in network traffic. Attention mechanisms further enhance detection by focusing on the most informative features and time steps. TSANet outputs a probability score for each traffic class, indicating whether the observed flow is benign or malicious. This hybrid architecture enables accurate detection while maintaining a lightweight model suitable for edge deployment as we can see TSANet Architecture in fig 2.

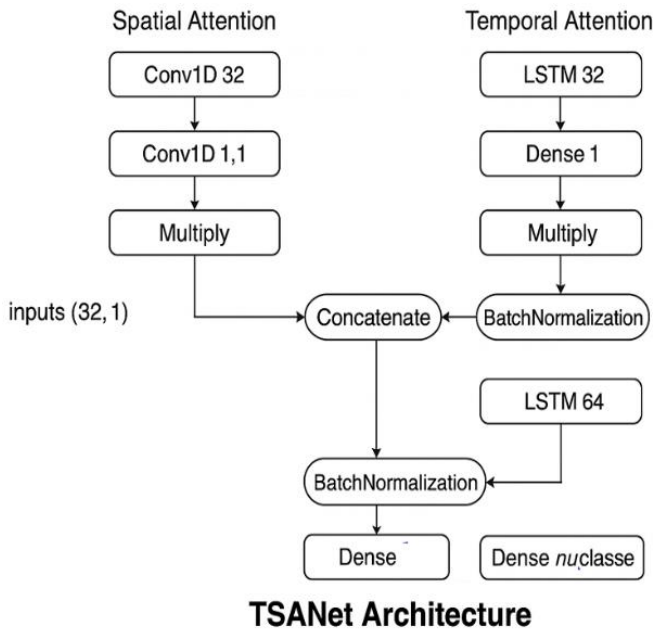

Fig 2: TSANet Architecture

3.1.5 Summary of System Flow

The complete system flow demonstrates a fully autonomous intrusion detection framework operating at the IoT edge. By combining real-time traffic monitoring, lightweight deep learning, and on-device inference, the proposed system achieves effective intrusion detection with low latency and minimal resource usage.

Chapter 4

Results

4.1 Offline Evaluation on CICIDS2017

The TSANet model was first evaluated offline using the CICIDS2017 dataset. The performance of the model was measured using standard classification metrics such as accuracy, precision, recall, and F1-score. The results show that the proposed TSANet model achieves an overall accuracy of **98.3%**, outperforming traditional machine learning and deep learning baseline models such as Random Forest, CNN, LSTM, and CNN–LSTM. The improvement in performance is mainly due to the combined spatial–temporal attention mechanism, which enables the model to focus on the most relevant features and time steps in network traffic. A comparison with baseline models demonstrates that while CNN and LSTM models perform well individually, their combination with attention mechanisms significantly enhances detection accuracy and robustness. Other Baseline Models are listed in Table 1.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	94.2	93.7	92.5	92.8
CNN	95.6	95.1	94.8	94.9
LSTM	96.2	96.0	95.7	95.8
CNN–LSTM	97.0	96.9	96.4	96.6
Proposed TSANet	**98.3**	**98.6**	**97.9**	**98.1**

Table 1: Comparison of TSANet with Baseline Models

4.2 Effect of Model Pruning

To make TSANet suitable for deployment on resource-constrained edge devices, structured pruning was applied to reduce model size and computational overhead. The pruning experiment evaluates the trade-off between compression and classification performance The results indicate that TSANet maintains stable accuracy even at moderate pruning levels. This shows that the model retains robustness after compression, making it suitable for real-time edge deployment without significant degradation in detection performance. For reference Figure 3 .

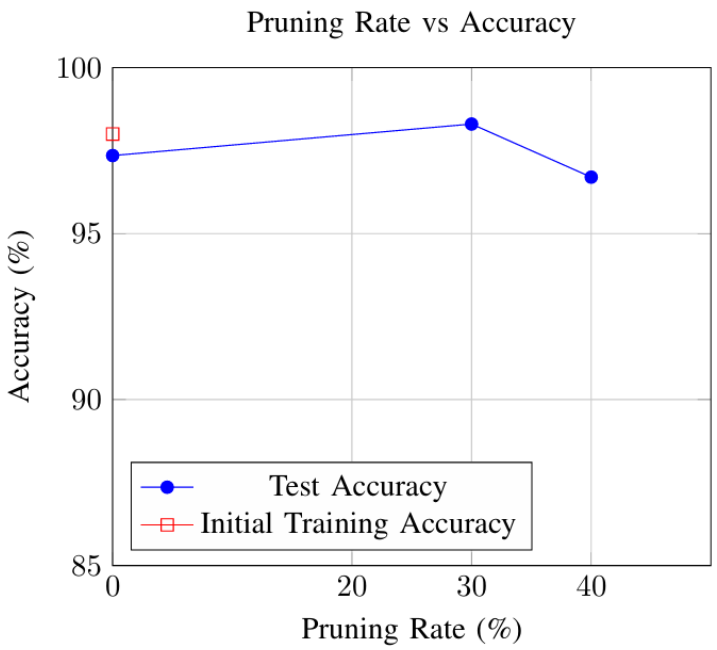

Fig 3: Impact of pruning rate on classification accuracy

4.3 Confusion Matrix Analysis

The confusion matrix analysis highlights the class-wise performance of the optimized TSANet model on the CICIDS2017 dataset. High detection rates are observed for volumetric attacks such as DoS and DDoS, indicating the effectiveness of the model in identifying high-impact network attacks Listed in table 2.

Actual / Predicted	Benign	Bot	BF	DDoS	DoS	Heartbleed	Infiltration
Benign	98.1	0.2	0.3	0.1	0.8	0.0	0.5
Bot	0.4	97.6	1.2	0.0	0.3	0.0	0.5
Brute Force	0.6	0.8	96.4	0.0	1.7	0.0	0.5
DDoS	0.0	0.0	0.0	98.9	1.1	0.0	0.0
DoS	0.4	0.1	0.3	1.5	97.3	0.1	0.3
Heartbleed	0.0	0.0	0.0	0.0	2.5	95.8	1.7
Infiltration	1.3	0.4	0.0	0.0	0.7	0.2	97.4

Table 2: Confusion Matrix of Optimized TSANet on CICIDS2017

4.4 Real-Time Deployment and Live Traffic Evaluation

To validate real-world applicability, the optimized TSANet model was deployed on a **Raspberry Pi 5**, configured as a standalone Wi-Fi access point. Multiple laptops and mobile devices were connected to the access point, generating genuine network traffic through activities such as web browsing, messaging, video streaming, system updates, and background processes.

Since all packets passed through the Raspberry Pi, complete bidirectional flows were captured, enabling accurate feature extraction and real-time inference. Each observed flow was identified by its source and destination IP addresses and ports, reconstructed into flow-level features, and classified by the TSANet model. During normal usage, traffic was correctly classified as **benign**, while attack traffic generated using controlled attack commands triggered **high-confidence alerts**. The system produced real-time outputs showing the source IP, destination IP, attack type, and prediction confidence.

Across several hours of continuous monitoring, the system achieved an average **real-time detection accuracy of 96.8%**. Detection latency remained within **8–12 milliseconds per flow**, and CPU and memory usage on the Raspberry Pi remained stable throughout operation.

These results confirm that deploying TSANet on a Raspberry Pi as a real network access point provides accurate packet timing, complete flow visibility, and reliable intrusion detection. The experiment validates the practicality of the proposed TSANet-based IDS for real-world IoT and edge computing environments.

The displayed output represents real-time alerts generated by the proposed TSANet-based intrusion detection system running on the Raspberry Pi. Each alert corresponds to a network flow observed at the access point, identified by the source IP address and port communicating with a destination IP address and port. For every such IP-to-IP flow, the system analyzes extracted traffic features and predicts whether the flow is **benign** or associated with a specific attack type (e.g., Heartbleed or DDoS), along with a confidence score indicating prediction certainty.

Normal user activities such as browsing or streaming are correctly classified as benign, while malicious traffic patterns trigger high-confidence attack alerts in real time.

```
===== ALERT =====
Flow: ('163.70.144.8', 443) → ('192.168.50.100', 45190)
Prediction: BENIGN (conf=0.5827)
=================

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: Heartbleed (conf=0.7354)
=================

===== ALERT =====
Flow: ('192.168.50.140', 52550) → ('34.107.221.82', 80)
Prediction: BENIGN (conf=0.5482)
=================

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: DDoS (conf=0.7714)
=================
```

Chapter 5

Conclusion and Future Scope

5.1 Conclusions and Future Work

This work presented TSANet, a lightweight Temporal–Spatial Attention Net work for real-time intrusion detection in IoT environments, designed for deploy ment on resource-constrained edge devices. By integrating spatial and temporal attention mechanisms with CNN and LSTM layers, the proposed system effec tively captures discriminative traffic features while maintaining low computa tional overhead. Experimental evaluation on the CICIDS2017 dataset demon strated superior detection performance compared to baseline models, achieving 98.3% offline accuracy. Real-time deployment on a Raspberry Pi configured as a Wi-Fi access point validated the practicality of the approach, with low in ference latency (8–12 ms per flow) and stable resource usage during live traffic monitoring. The system successfully detected both benign and malicious activ ities without reliance on cloud infrastructure. As future work, the model can be extended to support additional attack classes and adaptive learning for evolving threats, along with evaluation in large-scale distributed IoT environments

Bibliography

[1] United Nations, *Sustainable Development Goal 9: Industry, Innovation and Infrastructure*. Available: https://sdgs.un.org/goals/goal9, Accessed: 2025.

[2] I. Idrissi, M. Azizi, and O. Moussaoui, "A lightweight optimized deep learning-based host intrusion detection system deployed on the edge for IoT," *International Journal of Computing and Digital Systems*, vol. 11, no. 1, pp. 210–220, 2022.

[3] T. Saba, A. Rehman, T. Sadad, H. Kolivand, and S. A. Bahaj, "Anomaly-based intrusion detection system for IoT networks through deep learning model," *Computers & Electrical Engineering*, vol. 99, Article 107818, 2022.

[4] A. Gueriani, H. Kheddar, and A. C. Mazari, "Enhancing IoT security with CNN and LSTM-based intrusion detection systems," Technical Report, University of Medea, Algeria, 2023.

[5] Y. Otoum and A. Nayak, "DL-IDS: A deep learning-based intrusion detection framework for securing IoT," *Transactions on Emerging Telecommunications Technologies*, vol. 33, no. 3, e3803, 2022.

[6] B. F. Kayode, A. S. Akintola, O. Fagbohun, E. Anaesiuba, O. Ojumah, O. Odimayo, and P. C. Amajuoyi, "Temporal–Spatial Attention Network (TSANet) for DoS attack detection in network traffic," *arXiv preprint*, 2025.

[7] S. Racherla, P. Sripathi, N. Faruqui, M. A. Kabir, M. Whaiduzzaman, and S. A. Shah, "Deep IDS: A real-time intrusion detector for IoT nodes using deep learning," *IEEE Access*, vol. 12, pp. 63584–63598, 2024.

[8] R. Dahiya, M. Sharma, and M. Kumar, "Raspberry Pi IDS: A fruitful intrusion detection system for IoT," in *Proceedings of the 2nd International Conference on Innovative Practices in Technology and Management (ICIPTM)*, Noida, India, pp. 1–6, 2024.

[9] O. O. Olakanmi and S. O. Olatunji, "Optimized ensemble deep learning for real-time intrusion detection on resource-constrained Raspberry Pi devices," *Procedia Computer Science*, vol. 231, pp. 1991–1998, 2024.

[10] R. Adrian, R. D. Mandasari, and S. Alam, "Design and implementation of a machine learning-based adaptive IDS on Raspberry Pi for smart home network security," *SITEKIN: Jurnal Sains, Teknologi dan Industri*, vol. 22, no. 2, pp. 334–343, 2025.

[11] S. Tsimenidis, T. Lagkas, and K. Rantos, "Deep learning in IoT intrusion detection," *Journal of Network and Systems Management*, vol. 30, no. 8, pp. 1–40, 2022.

[12] R. Morshedi, S. M. Matinkhah, and M. T. Sadeghi, "Intrusion detection for IoT network security with deep learning," *Journal of Artificial Intelligence and Data Mining*, vol. 12, no. 1, pp. 37–55, 2024.

[13] D. Gupta, N. Yadav, S. Pande, and A. Khamparia, "Intrusion detection system on IoT with 5G network using deep learning," *Wireless Communications and Mobile Computing*, Article ID 1–13, 2022.

[14] N. Shone, T. N. Ngoc, V. D. Phai, and Q. Shi, "A deep learning approach to network intrusion detection," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 2, no. 1, pp. 41–50, 2018.